EFY TESTED
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Develop A Smart Android App **For Temperature Alert System**

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This project presents an Android app to generate an alert (alarm) on a smartphone when temperature of the process under consideration exceeds user-defined maximum temperature.

Alarm and annunciator instrumentation systems are widely used in the control room of process plants. When process variables being monitored by various on-field sensors exceed maximum/critical value, an alarm is generated on the PC or annunciator in the control room. This helps notify the operator to take necessary action to avoid any industrial disaster. This project is a smart temperature alert system based on a customised Android app, Arduino Uno and Bluetooth. The authors' prototype of the project is shown in Fig. 1.

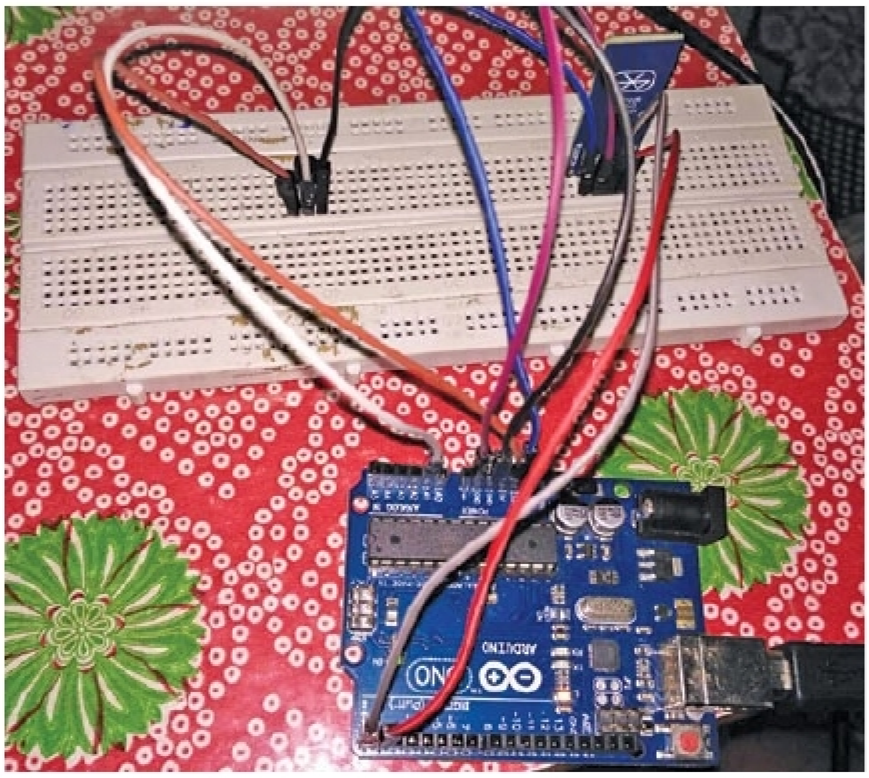

Fig. 1: Authors' prototype wired on breadboard

Circuit and working

The circuit diagram of the Android app for over-temperature alert is shown in Fig. 2. It is built around Arduino Uno board (Board1), Bluetooth module HC-05, temperature sensor LM35 and a few other components.

The system consists of a transmitter and a receiver unit. The circuit forms the transmitter unit, while the Android smartphone is the receiver unit. Temperature data, collected by LM35 IC temperature sensor from the field or surrounding area, is transmitted to the smartphone via Bluetooth. LM35 outputs analogue signal, and Arduino performs necessary voltage-to-temperature conversion before sending the bit pattern serially to Bluetooth module.

Components used in this project are detailed below.

LM35. This is a precision IC temperature sensor with its output voltage proportional to temperature (in °C). LM35 possesses low self-heating, and its measurement (sensing) range is from -55°C to 150°C.

In this project, pin 2 of LM35 is connected to analogue input pin A1 of Arduino. Power supply to LM35 is provided from 5V and GND pins of Arduino.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and fourteen digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM.

In this project, Arduino is used to convert output voltage of LM35 into a corresponding temperature

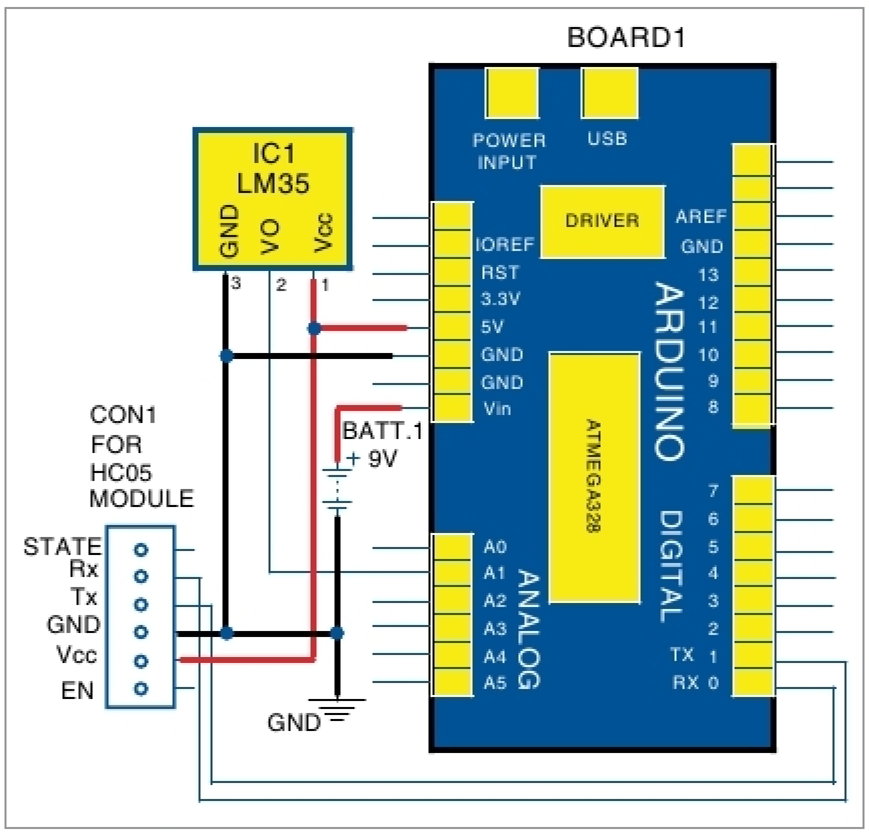

Fig. 2: Circuit diagram of the Android app for temperature alert

Fig. 3: Screenshot of temperature data as observed on the smartphone